Quiz 7

1 Grading Scheme

- Question 1 is for 3 points (1 point for base case, 1 point for Induction hypothesis and 1 point for the proof)
- Question 2 is for 2 points (1 point for correctly using the formulae and 1 point for solving for c_1, c_0 correctly)
- Question 3 is for 2 points (1 point for correctly computing $\Delta^i(a_0)$ and 1 point for finding the correct polynomial)

Total grade is out of 7.

2 Solutions

1. Base case: n = 4. As $2^4 = 16 < 24 = 4!$, the base case holds.

   Induction hypothesis: Let $k \geq 4$ and suppose that $2^k < k!$.

   Induction step: $2^{k+1} = 2 \times 2^k < 2 \times k! < (k+1) \times k! = (k+1)!$. Therefore, $2^{k+1} < (k+1)!$.

2. $a_n = 2a_{n-1} + 1$. Using Prop 23.1, we know that $a_n = c_1 2^n + c_0$.

$$1 = a_1 = c_1 + c_0$$

$$3 = a_2 = 2c_1 + c_0$$

   Solving for c_0 and c_1, we get $c_0 = -1$ and $c_1 = 2$. Therefore, $a_n = 2^{n+1} - 1$

3. $a_0 = 1$, $a_1 = 2$, $a_2 = 4$, $a_3 = 8$, $a_4 = 15$. Note that $\Delta(a_0) = 1$, $\Delta^2(a_0) = 1$, $\Delta^3(a_0) = 1$ and $\Delta^{i \geq 4}(a_0) = 0$.

   Using the formulae, we get

$$\begin{aligned} p(x) &= a_0\binom{n}{0} + \Delta(a_0)\binom{n}{1} + \Delta^2(a_0)\binom{n}{2} + \Delta^3(a_0)\binom{n}{3} \\ &= 1 + n + n(n-1)/2 + n(n-1)(n-2)/6 \end{aligned}$$

3 Common error

1. Write the induction hypothesis clearly.
2. $(a+b)! \neq a! + b!$ and $a(b!) \neq (ab)!$.